CBCS SCHEME

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BMATE201

Second Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Mathematics-II for EEE Stream

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Find the directional derivative of $\phi = xy^3 + yz^3$ at (2, -1, 1) along $i + 2j + 2k$.	7	L2	CO1
	b.	Find div $\vec{F}$ and curl $\vec{F}$, if $\vec{F} = y^3z^2\hat{i} + 3xy^2z^2\hat{j} + 2xy^3z\hat{k}$ at (1, -1, 1).	7	L2	CO1
	c.	If $\vec{V} = 3xy^2z^2\hat{i} + y^3z^2\hat{j} + 2y^2z^3\hat{k}$ and $\vec{F} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}$, show that $\vec{V}$ is solenoidal and $\vec{F}$ is irrotational.	6	L2	CO1
		OR			
Q.2	a.	Find the workdone in moving a particle by the force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the straight line from (0, 0, 0) to (2, 1, 3).	7	L2	CO1
	b.	Using Green's theorem evaluate $\oint (xy - x^2)dx + x^2y\,dy$ over the region bounded by $y = x$ and $y = x^2$.	7	L3	CO1
	c.	Write the mathematical tool program to find the divergence of the vector field. $\vec{F} = x^2yz\hat{i} + y^2zx\hat{j} + z^2xy\hat{k}$.	6	L3	CO5
		Module – 2			
Q.3	a.	If W is the set of all points in R^3 satisfying the equation $a_1x_1 + a_2x_2 + a_3x_3 = 0$, then prove that W is a subspace of R^3.	7	L2	CO2
	b.	Prove that in $V_3(R)$, the vectors {(1, 2, 1), (2, 1, 0), (1, -1, 2)} are linearly independent.	7	L2	CO2
	c.	Prove that the transformation $T : R^3 \to R^3$ defined by $T(x_1, x_2, x_3) = (0, x_2, x_3)$ is linear.	6	L2	CO2
		OR			
Q.4	a.	Express the vector (2, -1, -8) as a linear combination of the vectors (1, 2, 1), (1, 1, -1) and (4, 5, -2).	7	L2	CO2
	b.	Find the matrix of the linear transformation $T : V_3(R) \to V_2(R)$ defined by $T(x, y, z) = (x + y, y + z)$ relative to the bases $B_1 = \{(1, 1, 0), (1, 0, 1), (1, 1, -1)\}$, $B_2 = \{(2, -3), (1, 4)\}$.	7	L2	CO2

<table>
<tr><td></td><td>c.</td><td>Using the modern mathematical tool. Write the code to represent the reflection transformation $T : R^2 \rightarrow R^2$ and to find the image of vector (10, 0) when it is reflected about the y-axis.</td><td>6</td><td>L3</td><td>CO5</td></tr>
<tr><td colspan="6">Module – 3</td></tr>
<tr><td>Q.5</td><td>a.</td><td>Find the Laplace transform of i) $te^{-t}\sin 3t$ ii) $\dfrac{e^{at} - e^{bt}}{t}$</td><td>7</td><td>L2</td><td>CO3</td></tr>
<tr><td></td><td>b.</td><td>Find the Laplace transform of the square wave function of period 'a' defined by
$f(t) = \begin{cases} E, & 0 < t < a/2 \\ -E, & a/2 < t < a \end{cases}$</td><td>7</td><td>L3</td><td>CO3</td></tr>
<tr><td></td><td>c.</td><td>Express $f(t) = \begin{cases} t^2, & 0 < t < 2 \\ 4t, & 2 < t < 4 \\ 6, & t > 4 \end{cases}$ interms of unit step function and hence find its Laplace transform.</td><td>6</td><td>L3</td><td>CO3</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>Q.6</td><td>a.</td><td>Find $L^{-1}\left\{\dfrac{2s-1}{(s-1)(s-3)}\right\}$.</td><td>7</td><td>L2</td><td>CO3</td></tr>
<tr><td></td><td>b.</td><td>Find $L^{-1}\left\{\dfrac{1}{s(s^2+a^2)}\right\}$ using convolution theorem.</td><td>7</td><td>L2</td><td>CO3</td></tr>
<tr><td></td><td>c.</td><td>Solve by Laplace transform $\dfrac{d^2y}{dt^2} - \dfrac{dy}{dt} = 0$, $y(0) = y'(0) = 3$.</td><td>6</td><td>L3</td><td>CO3</td></tr>
<tr><td colspan="6">Module – 4</td></tr>
<tr><td>Q.7</td><td>a.</td><td>Find the real root of the equation $x \sin x + \cos x = 0$ near $x = \pi$ by Newton-Raphson method.</td><td>7</td><td>L2</td><td>CO4</td></tr>
<tr><td></td><td>b.</td><td>Using Newton's forward interpolation formula find y at x = 5 for the data:
<table><tr><td>x</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>y</td><td>1</td><td>3</td><td>8</td><td>16</td></tr></table></td><td>7</td><td>L3</td><td>CO4</td></tr>
<tr><td></td><td>c.</td><td>Find the interpolating polynomial using Newton's divided difference formula for the data:
<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>5</td></tr><tr><td>y</td><td>2</td><td>3</td><td>12</td><td>147</td></tr></table></td><td>6</td><td>L2</td><td>CO4</td></tr>
<tr><td colspan="6">OR</td></tr>
<tr><td>Q.8</td><td>a.</td><td>Find the real root of the equation $x\log_{10}x = 1.2$ by Regula Falsi method between 2 and 3 (three iterations).</td><td>7</td><td>L2</td><td>CO4</td></tr>
<tr><td></td><td>b.</td><td>Find y at x = 5, if y(1) = +3, y(3) = 9, y(4) = 30, y(6) = 132 using Lagrange's interpolation formula.</td><td>7</td><td>L2</td><td>CO4</td></tr>
</table>

			Marks	Level	CO
	c.	Evaluate $\int_0^{\pi/2} \sqrt{\cos x}\, dx$ by using Simpson's $\left(\frac{1}{3}\right)^{rd}$ rule by taking 6 equal parts.	6	L3	CO4
		Module – 5			
Q.9	a.	Use Taylor's series method to find y(0.1) by considering upto 3rd degree term, given $\frac{dy}{dx} = x - y^2$, y(0) = 1.	7	L3	CO4
	b.	Given $\frac{dy}{dx} = 3e^x + 2y$, y(0) = 0, h = 0.1 using Runge-Kutta method of 4th order find y at x = 0.1.	7	L3	CO4
	c.	Apply Milne's predictor-corrector method, find y(0.4) given $\frac{dy}{dx} = x^2 + y^2$, y(0) = 1, y(0.1) = 1.1113, y(0.2) = 1.2507, y(0.3) = 1.426.	6	L2	CO4
		OR			
Q.10	a.	Using modified Euler's method, find y(0.1) given that $\frac{dy}{dx} = x^2 + y$ and y(0) = 1, take h = 0.05 and perform two iterations in each stage.	7	L2	CO4
	b.	Apply Milne's method to find y(4.4) given that y(4) = 1, y(4.1) = 1.0049, y(4.2) = 1.0097, y(4.3) = 1.0142 and $\frac{dy}{dx} = \frac{2 - y^2}{5x}$.	7	L2	CO4
	c.	Write a modern mathematical tool program to solve $\frac{dy}{dx} = x^2 + y^2$, y(0) = 1, h = 0.1 using R – K method of 4th order.	6	L3	CO5

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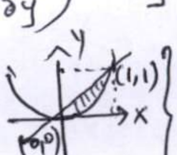
13/9/23 13/9/23
Signature of Scrutinizer

Scheme & Solutions

Subject Title : Mathematics II for EES **Subject Code :** BMATE 201

Question Number	Solution	Marks Allocated
Q.01 (a)	$\nabla\phi = y^3 i + (3xy^2+z^3) j + (3yz^2 k)$, at $(2,-1,1)$ $\nabla\phi = -i+7j-3k$	3+1=4
	$\hat{n} = \frac{i+2j+2k}{3}$, $\nabla\phi\cdot\hat{n} = 7/3$	1+2=3
(b)	$\nabla\cdot\vec{F} = 0 + 6xyz^2 + 2xy^3$ at $(1,-1,1)$ $\nabla\cdot\vec{F} = -8$ ----	3
	Curl $\vec{F} = \nabla\times\vec{F} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^3z^2 & 3xy^2z^2 & 2xy^3z \end{vmatrix} = 0$, on expansion	1+3
(c)	$\operatorname{div}\vec{v} = \nabla\cdot\vec{v} = 3y^2z^2 + 3y^2z^2 - 6y^2z^2 = 0 \Rightarrow \vec{v}$ is solenoidal	3
	curl $\vec{F} = \nabla\times\vec{F} = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2-yz & y^2-zx & z^2-xy \end{vmatrix}$ on expansion $\nabla\times\vec{F} = 0$ $\Rightarrow \vec{F}$ is irrotational	3
Q.02 (a)	equⁿ. of the st line from $(0,0,0)$ to $(2,1,3)$ $\frac{x}{2} = \frac{y}{1} = \frac{z}{3} = t$, $\therefore x = 2t,\ y = t,\ z = 3t$. t Varies from 0 to 1	2
	workdone $= \int_C \vec{F}\cdot d\vec{r} = \int_0^1 (36t^2 + 8t)\,dt = 16$	5
(b)	by Green's theorem $\oint_C M\,dx + N\,dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy$	1
	$M = xy - x^2$, $\frac{\partial M}{\partial y} = x$, $N = x^2y$, $\frac{\partial N}{\partial x} = 2xy$ 	2
	$\therefore \oint_C (xy - x^2)\,dx + x^2y\,dy = \int_{x=0}^{1}\int_{y=x^2}^{x} (2xy - x)\,dy\,dx$ $= \int_{x=0}^{1} \left[2xy^2/2 - xy\right]_{x^2}^{x} dx$ getting $\oint_C (xy - x^2)\,dx + x^2y\,dy = 0$	3

Subject Title : Matrematics-II for EES   Subject Code : BMATE 201

Question Number	Solution	Marks Allocated

(c)

```
from sympy.vector import *
from sympy import symbols
N = CoordSys3D ('N')
x, y, z = symbols ('x, y, z')
A= N.x**2*N.y*N.z*N.i+N.y**2*N.z*N.x*N.j+N.z**2*N.
x*N.y*N.k
delop = Del()
divA = delop.dot(A)
Print (divA)
Print (f"\n Divergence of {A} is \n")
Print (divergence(A))
```

(3) (3)

Q.03

a) Let $W = \{ x_1, x_2, x_3 / a_1x_1 + a_2x_2 + a_3x_3 = 0 \}$

Let $\alpha = (x_1, x_2, x_3)$, $\beta = (y_1, y_2, y_3) \in W$

$\ni a_1x_1 + a_2x_2 + a_3x_3 = 0$, $a_1y_1 + a_2y_2 + a_3y_3 = 0$

$\alpha + \beta = (x_1 + y_1, x_2 + y_2, x_3 + y_3)$

Consider $a_1(x_1+y_1) + a_2(x_2+y_2) + a_3(x_3+y_3)$

rearranging we get $\alpha + \beta = 0$ closed under addition (5)

III'y $\alpha \in W$, $c_1 \in F$

Proving $c_1\alpha = c_1(x_1, x_2, x_3)$,
$= (c_1x_1, c_2x_2, c_3x_3)$ (2)

Consider $a_1c_1x_1 + a_2c_2x_2 + a_3c_3x_3 = c_1(a_1x_1 + a_2x_2 + a_3x_3) = c_1(0) = 0$, $\therefore$ W is a subspace of

b) Let $\lambda_1(1,2,1) + \lambda_2(2,1,0) + \lambda_3(1,-1,2) = (0,0,0)$ (1)

getting $\lambda_1 + 2\lambda_1 + \lambda_3 = 0$, $2\lambda_1 + \lambda_2 - \lambda_3 = 0$,

$\lambda_1 + 2\lambda_3 = 0$ (3)

and $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 0 & 2 \end{bmatrix}$ $\rightarrow$ Reduce it to row

echelon form getting $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & -3 \\ 0 & 0 & 12 \end{bmatrix}$ (2)

$\therefore$ Rank = 3 = no of unknowns, system of equations are homogeneous $\therefore$ $\lambda_1 = 0$, $\lambda_2 = 0$, $\lambda_3 = 0$ (1)

$\Rightarrow$ given vectors are linearly independent

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<table>
<tr><th>Question Number</th><th>Solution</th><th>Marks Allocated</th></tr>
<tr><td>(c)</td><td>$\alpha = (x_1, x_2, x_3)$, $\beta(y_1, y_2, y_3) \in V_3(R)$
$T(\alpha+\beta) = T[(x_1, x_2, x_3) + (y_1, y_2, y_3)] = T[(x_1+y_1), x_2+y_2, x_3+y_3]$
$= [0, x_2+y_2, x_3+y_3]$
$= [0, x_2, x_3] + [0, y_2, y_3]$
$= T(\alpha) + T(\beta)$
$T(c\alpha) = T[c(x_1, x_2, x_3)]$, $c \in R$
$= (0, cx_2, cx_3) = cT(\alpha)$</td><td>3

3</td></tr>
<tr><td>Q.04
(a)</td><td>$(2,-1,8) = c_1(1,2,1) + c_2(1,1,-1) + c_3(4,5,-2)$
$c_1 + c_2 + 4c_3 = 2$, $2c_1 + c_2 + 5c_3 = -1$, $c_1 - c_2 - 2c_3 = 8$ →
solving these getting $c_1 = -\frac{13}{3}$, $c_2 = 1$, $c_3 = \frac{4}{3}$</td><td>1
2
4</td></tr>
<tr><td>(b)</td><td>$T(1,1,0) = (2,1)$, $T(1,0,1) = (1,1)$, $T(1,1,-1) = (2,0)$
expressing $(2,1)$, $(1,1)$ and $(2,0)$ as a linear Combination of $(2,-3)$ and $(1,4)$
getting in each case $c_1 = \frac{7}{11}$, $c_2 = \frac{8}{11}$
$c_1 = \frac{3}{11}$, $c_2 = 5/11$, $c_1 = \frac{8}{11}$, $c_2 = \frac{6}{11}$
and $A_T = \begin{bmatrix} \frac{7}{11} & \frac{3}{11} & 8/11 \\ \frac{8}{11} & 5/11 & 6/11 \end{bmatrix}$</td><td>2

5</td></tr>
<tr><td>(c)

Book work</td><td>$\{(1,0,0), (0,1,0)(0,0,1)\}$ is a basis of $V_3(R)$, $T(1,0,0) = (1,1,2) = \alpha_1$
$T(0,1,0) = (1,-1,0) = \alpha_2$, $T(0,0,1) = (0,0,1) = \alpha_3$
$R(t)$ is a subspace spanned by $(\alpha_1, \alpha_2, \alpha_3)$.
$c_1(1,1,2) + c_2(1,-1,0) + c_3(0,0,1) = 0 \Rightarrow c_1 = 0, c_2 = 0, c_3 = 0$
$\therefore (\alpha_1, \alpha_2, \alpha_3)$ is L.I and hence a basis of $R(t)$, $\therefore r(T) = 3$
Let $(x,y,z) \in N(T) \Rightarrow T(x,y,z) = 0 \Rightarrow (x+y, x-y, 2x+z) = 0$
$\Rightarrow x = 0, y = 0, z = 0$, $\therefore N(T) = \{(0,0,0)\}$ $\therefore n(T) = 0$
rank + nullity = d(V) hence verified.
$3 + 0 = 3$</td><td>2

2

2

6</td></tr>
</table>

Subject Title : Mathematics - II for EES Subject Code : BMAT201

Question Number	Solution	Marks Allocated
Q.No 5 (a)	(i) $L\left[e^{-t}\sin 3t\right] = \frac{3}{(s+1)^2+9}$, $L\left[t e^{-t}\sin 3t\right] = \frac{6(s+1)}{\left[(s+1)^2+9\right]^2}$	3
	(ii) $L\left\{\frac{e^{at}-e^{bt}}{t}\right\} = \int_s^\infty \left(\frac{1}{s-a} - \frac{1}{s-b}\right)ds = \log\left(\frac{s-a}{s-b}\right)_s^\infty$ $= \log\left(\frac{s-b}{s-a}\right)$	1 + 1 + 1 + 1
(b)	$L\{f(t)\} = \frac{1}{1-e^{-sT}}\int_0^T e^{-st} f(t)dt = \frac{1}{1-e^{-sT}}\left\{\int_0^{a/2} E e^{-st}dt + \int_{a/2}^{a} -E\, e^{-st}\, dt\right\}$	1
	$= \frac{E}{1-e^{-as}}\left\{\left[\frac{e^{-st}}{-s}\right]_0^{a/2} \mp \left[\frac{e^{-st}}{-s}\right]_{a/2}^{a}\right\}$	2
	getting $L\{f(t)\} = \frac{E\,(1-e^{-as/2})}{s\,(1+e^{-as/2})}$	4
(c)	$f(t) = t^2 + (4t-t^2)u(t-2) + (6-4t)u(t-4)$	1
	$f(t) = t^2 + \left[4(t+2) - (t+2)^2\right]u(t-2) + \left[6-4(t+4)\right]u(t-4)$	2
	getting $L\{f(t)\} = \frac{2}{s^3} + e^{-2s}\left[\frac{4}{s} - \frac{2}{s^3}\right] + e^{-4s}\left(\frac{-4}{s^2} - \frac{10}{s}\right)$	4
Q.No 6 (a)	$\frac{2s-1}{(s-1)(s-3)} = \frac{A}{s-1} + \frac{B}{s-3}$	1
	getting $A = -1/2$, $B = 5/2$	4
	$L\left\{\frac{2s-1}{(s-1)(s-3)}\right\} = -\frac{1}{2}e^{t} + \frac{5}{2}e^{3t}$	2
(b)	$f(t)=1$, $g(t) = \frac{1}{a}\sin at$	2
	$L^{-1}\left[F(s)\cdot G(s)\right] = \int_{u=0}^{t} f(u)g(t-u)du = \int_{u=0}^{t}\frac{1}{a}\sin(at-au)du$	2

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Question Number	Solution	Marks Allocated
	getting $L^{-1}\left[\frac{1}{s(s^2+a^2)}\right] = \frac{1}{a^2}\left[1-\cos at\right]$	3
(c)	$s^2 L\{y(t)\} - s y(0) - y'(0) - [s L(y(t)) - y(0) = 0$	2
	getting $L\{y(t)\} = \frac{3s}{s^2-s}$	3
	$\therefore y(t) = 3e^t$	1
Q.No 7 (a)	$f'(x) = x\cos x$, $x_1 = 2.8233$, $x_2 = 2.7986$	1+2+2
	$x_3 = 2.7984$, $\cancel{x_4 = 2.7984}$	+2
	$\boxed{\text{root} = 2.7984}$	
(b)	x y Δy $\Delta^2 y$ $\Delta^3 y$	
	4 1 2	
	6 3 3	3
	8 8 5 0	
	3 16 8 3	
	$r = 0.5$, $y(5) = 1.1825$	4
(c)	$f(x_0, x_1) = 1$, $f(x_1, x_2) = 9$, $f(x_2, x_3) = 45$	
	$f(x_0, x_1, x_2) = 4$, $f(x_1, x_2, x_3) = 9$, $f(x_0, x_1, x_2, x_3) = 1$	4
	getting $f(x) = x^3 + x^2 - x + 2$	2
Q.No 8 (a)	$x = \frac{a f(b) - b f(a)}{f(b) - f(a)}$, $a = 2$, $b = 3$, $f(2) = -0.59794$, $f(3) = 0.23136$	
	$x_1 = 2.72102$, $f(x_1) = -0.01709$ root in $(2.72012, 3)$	2
	$x_2 = 2.74021$, $f(x_2) = -0.0003803$ root in $(2.74021, 3)$	2
	$x_3 = 2.74024$.	2
	required root 2.74024	$\cancel{2}$

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Question Number	Solution	Marks Allocated
(b)	$y(5) = \frac{(5-3)(5-4)(5-6)(3)}{(1-3)(1-4)(1-6)} + \frac{(5-1)(5-4)(5-6)(9)}{(3-1)(3-4)(3-6)}$ $+ \frac{(5-1)(5-3)(5-6)(30)}{(4-1)(4-3)(4-6)} + \frac{(5-1)(5-3)(5-4)(132)}{(6-1)(6-3)(6-5)}$ getting $y(5) = 64.9$	4 3
(c)	$h = \pi/12$ x: 0, $\pi/12$, $\pi/6$, $\pi/4$, $\pi/3$, $5\pi/12$, $\pi/2$ y: 1, 0.9828, 0.9306, 0.8409, 0.7071, 0.5087, 0 getting $I = 1.1873$	4 2
Q.No 9 (a)	$x_0 = 0,\ y_0 = 1,\ y' = x - y^2,\ y'(0) = -1$ $y''(0) = 3,\ y'''(0) = -8$ getting $y(0.1) = 1 - 0.1 + 0.0015 + 0.00133$ $= 0.090017$.	1 4 2
(b)	$f(x,y) = 3e^x + 2y,\ x_0 = 0,\ y_0 = 0,\ h = 0.1$ $K_1 = 0.3,\ K=2 = 0.3454,\ K_3 = 0.3499$ $K_4 = 0.4015$ getting $y(0.1) = 0.3487$	5 2
(c)	$y_0' = 1,\ y_1' = 1.245,\ y_2' = 1.6043,\ y_3' = 2.1235$ $y_4^{(P)} = 1.6844$ $y_4' = 2.9972$ $y_4^{(C)} = 1.6872$ $\therefore y(0.4) = 1.6872$	3 3

Subject Title : Matematics - II for EES **Subject Code :** BMATE201

Question Number	Solution	Marks Allocated

Q.10 (a)

$x_0 = 1, \; y_0 = 1, \; f(x,y) = x^2 + y, \; h = 0.05,$

$y_1^{(0)} = 1.05$ (2)

I Stage : $y_1^{(1)} = 1.0513, \quad y_1^{(2)} = 1.0513$ (3)

II Stage : $y_1^{(0)} = 1.104, \quad y_1^{(1)} = 1.1055 = y_1^{(2)} \therefore y(0.1) = 1.1055$ (3)

(b)

$y_0' = 0.05, \quad y_1' = 0.0483, \quad y_2' = 0.0467,$

$y_3' = 0.0452$ (4)

$y_4^{(P)} = 1.0187, \quad y_4^{(1)} = 0.0437$

$y_4^{(C)} = 1.0186$ (3)

$\therefore \; y(0.4) = 1.0186$

(c)

```
def f(x,y):
    return x*2 + y*2

def runge_kutta(h, x0, y0, x):
    n=int((x-x0/h)
    y = y0
    for _ in range(n):
        k1 = h*f (x0,y)
        k2 = h*f (x0+h/2, y+k1/2)
        k3 = h*f (x0+h/2, y+k2/2)
        k4 = h*f(x0+h, y+k3)
        y+ = (k1+2*k2+2*k3+k4)/6
        x0+ = h
    return y

x0 =0
y0 =1
x =0.1
h =0.1
result = runge_kutta(h, x0, y0, x)
Print("y(0.1)=", result)
```

Marks: (2), (3), (1)